Entry Without Displacement: Open-Weight Models and the Market for Inference

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Abstract

Open-weight models are widely expected to constrain the pricing power of proprietary model providers, but the mechanism has not been measured. We assemble a market-wide panel of the models served by a large inference router, covering posted prices, capability benchmarks, weight-release status and revealed usage for 457 models and extended back to 2023 with archived catalogues. Competition works through releases rather than repricing. 54% of models are never repriced at all, and a posted price changes in only 12.6% of model-months, even though changing it costs a provider nothing. When prices do change, volume responds weakly. The own-price elasticity is -0.53 (s.e. 0.10), identified from 708 revisions with model and date fixed effects. And entry does not displace incumbents. Over four weeks, models released inside the window took 28% of tokens while incumbent volume grew 18%, and we can rule out differential losses above 0.13 log points for the incumbents closest to an entrant in capability space. Removing open-weight models from the choice set in an estimated nested-logit demand system raises the average price paid per token by 156% and costs users about \$2.4m per day on this platform, while raising provider concentration. Open weights discipline the price at which new demand is served, not the prices incumbents post.

1 Introduction

Whether open-weight models constrain the market power of proprietary providers is now a policy question as much as an economic one. Reports commissioned by governments name it as an open evidence gap [Bengio et al., 2026], and the theoretical case cuts both ways. Scale economies in training push toward concentration [Kaplan et al., 2020, Korinek and Vipra, 2025], while downloadable weights supply a competitive fringe that any host can serve, the mechanism open-source software has long been argued to supply in adjacent markets [Lerner and Tirole, 2002, Nagle, 2019]. Demirer et al. [2025] document a collapse in the price of a unit of measured intelligence and a proliferation of suppliers, and Nagle and Yue [2025] show that much proprietary usage sits on models dominated on price and measured capability by an open alternative.

Both results describe the allocation of demand across models at a point in time. Neither identifies the channel through which open weights would constrain proprietary pricing, and the two candidates have opposite implications for policy. If open weights force incumbents to cut posted prices, the constraint is visible in the price series of established models. If instead open weights capture new demand while incumbents hold their prices, the constraint operates on the terms at which the market grows, and any measurement built on incumbent price responses will find nothing.

We assemble a panel that separates the two, combining the current cross-section of a large inference router (457 models with observed usage, 175 of them open-weight), a daily usage panel covering all of them for one month, task-level shares for 29 uses, per-provider hosting terms, and archived price catalogues and leaderboards reaching back to 2023. The panel used for price responses contains 19,803 model-date observations on 634 models. Three results follow, and they point the same way. First, the posted price of a given model version behaves like a fixed characteristic of the product. 54%

of models are never repriced, and the monthly frequency of a price change is 12.6%, comparable to consumer goods [Nakamura and Steinsson, 2008, Klenow and Malin, 2011] in a setting where the menu cost that explains rigidity there does not exist. Second, when prices do change, volume moves little. Exploiting 708 revisions with model and date fixed effects, we estimate an own-price elasticity of -0.53 (s.e. 0.10), below the value near one reported by Demirer et al. [2025] and far below what homogeneous-goods reasoning would imply. Third, entry is absorbed by growth rather than by displacement. Over the four weeks of the daily panel the market grew 65%, models released inside the window took 28% of tokens, and incumbent volume still rose 18%. In an incumbent-by-entrant event study we find no differential decline for the incumbents technologically closest to an entrant.

We then quantify the constraint where it does operate. We estimate a nested-logit demand system over the 172 models with prices and capability scores, invert it to recover mean utilities that reproduce observed shares, and delete open-weight models from the choice set. Users would pay 156% more per token, and the surplus loss is worth about \$2.4m per day on this platform alone. Across 29 task-level markets the implied price increase rises almost one for one with the open-weight share of task spending. Concentration moves against the usual presumption, since removing open weights raises the provider Herfindahl index from 897 to 1048.

Our contribution is to identify the margin on which open-weight competition operates and to price it. The measurement of price rigidity and of the within-model demand response is, to our knowledge, new to this market, and it explains why the reallocation gains documented by Nagle and Yue [2025] persist. With inelastic short-run demand a provider has little to gain from cutting the price of a model already in the field, and users have little reason to move quickly. We release the collection code, which reconstructs the entire panel from public endpoints.

2 Data

2.1 Sources

We use the public endpoints of OpenRouter, a router that serves requests to hundreds of models from many providers through one interface. Four supply the raw material. A catalogue gives posted prices, context length, release timestamp and the Hugging Face repository for models whose weights are published. A leaderboard gives tokens, requests and tool calls by model, a benchmark table gives third-party capability scores, and a task table gives each model's share of spending in 29 usage categories. Each model's own page embeds a daily usage series for the trailing month, which we collect for every model with observed usage, and per-provider hosting terms come from the endpoint listing. Historical coverage comes from Internet Archive captures of the catalogue and of the leaderboard, the latter embedding the leaderboard payload in the page's streamed render data. Appendix A documents each step.

2.2 Openness

We classify model j as open-weight, $o_j = 1$, if the catalogue records a public weight repository for it. The classification is model-specific rather than provider-specific, which matters here because several providers, among them Google, OpenAI and Alibaba, ship both open-weight and proprietary models, and a provider-level rule would misclassify them. The definition follows the one used in the policy literature, where the salient feature is that weights can be downloaded and served by a third party [Kapoor et al., 2024, NTIA, 2024].

2.3 Capability

Capability enters as a vector $q_j \in \mathbb{R}^M$ of third-party scores over M dimensions covering reasoning, coding and agentic use, supplemented by pairwise-comparison scores from a design-oriented arena in the spirit of Chiang et al. [2024]. These measures are highly collinear. Writing $\bar{q}_m$ and ς_m for the mean and standard deviation of dimension m across models, and Λ for the covariance matrix of the standardised scores, the leading eigenvector w of Λ defines the capability index

$$Q_j = w'\tilde{q}_j, \qquad \tilde{q}_{jm} = (q_{jm} - \bar{q}_m)/\varsigma_m, \tag{1}$$

which absorbs 98% of the variance of the three benchmark dimensions and 91% when the two arena blocks are added. Where a dimension is missing we impute it by ridge regression on the dimensions a model does have, recovering held-out values with R^2 above 0.96 and giving 180 scored models covering 91% of tokens.

Where a pairwise notion of substitutability is needed we use the whitened distance between capability vectors and the similarity it induces,

$$d_{jk} = \sqrt{(\tilde{q}_j - \tilde{q}_k)'\Lambda^{-1}(\tilde{q}_j - \tilde{q}_k)}, \qquad S_{jk} = \exp(-\kappa\, d_{jk}), \tag{2}$$

with Λ ridge-shrunk and κ normalised so that the median pair maps to e^{-1}. Section 5 reports results under Euclidean and cosine alternatives.

2.4 Prices and quantities

Providers post separate input and output prices, p^{in}_{jt} and p^{out}_{jt}, and models differ sharply in how many output tokens they emit per input token. We therefore work with the blended price

$$p_{jt} = \omega_{jt}\, p^{\text{in}}_{jt} + (1 - \omega_{jt})\, p^{\text{out}}_{jt}, \qquad \omega_{jt} = \frac{I_{jt}}{I_{jt} + O_{jt}}, \tag{3}$$

where I_{jt} and O_{jt} are input and output tokens. For the panel regressions we replace ω_{jt} by its sample mean, so the index carries no feedback from a model's own usage composition. Quantities are tokens, $y_{jt} = I_{jt} + O_{jt}$, and spending is $e_{jt} = (p^{\text{in}}_{jt} I_{jt} + p^{\text{out}}_{jt} O_{jt})/10^6$.

2.5 Sample and market definition

Table 1 summarises the estimation sample. The platform is one segment of the market for inference, excluding traffic sent directly to a provider's own API and consumer subscription products, and its users are developers who have chosen a router. That selects for price sensitivity and willingness to switch, so the segment we observe is the one in which open-weight substitution is easiest.

Table 1: The estimation sample, measured on the last complete day of the daily panel. Prices are dollars per million tokens for the blended mix of (3). Capability is the index (1). Shares are within the estimation sample, which covers 91% of platform tokens.

	Models	Median price	Median capability	Median context (thousands)	Token share	Spend share
Open-weight	86	0.29	30.5	256	0.73	0.32
Proprietary	86	1.50	36.6	400	0.27	0.68
All scored	172	0.59	33.4	262	1.00	1.00

3 Market structure

3.1 Prices and capability

Figure 1 places every model with usage on the price-capability plane. At any capability level below the frontier, prices differ by one to two orders of magnitude, so posted price is far from a sufficient statistic for what users buy, and the open-weight side is systematically cheaper. Matching each proprietary model to the open-weight models within two points of it on the capability index, the median proprietary model costs 3.3 times the median open alternative, and 1.7 times weighted by volume. Averaged over usage a token from an open-weight model costs \$0.35 per million against \$1.95. These gaps are smaller than the ninety percent difference reported for 2025 data [Demirer et al., 2025, Nagle and Yue, 2025], as one would expect if proprietary providers have since released cheaper tiers.

The right panel tracks the best capability obtainable from each side over time. The open-weight frontier reaches a given capability level a median of 105 days after the proprietary frontier first reaches it. The gap is a lag rather than a level. At the very top of the capability distribution proprietary models are unmatched at every date, but the set of tasks for which that matters shrinks as the open frontier advances.

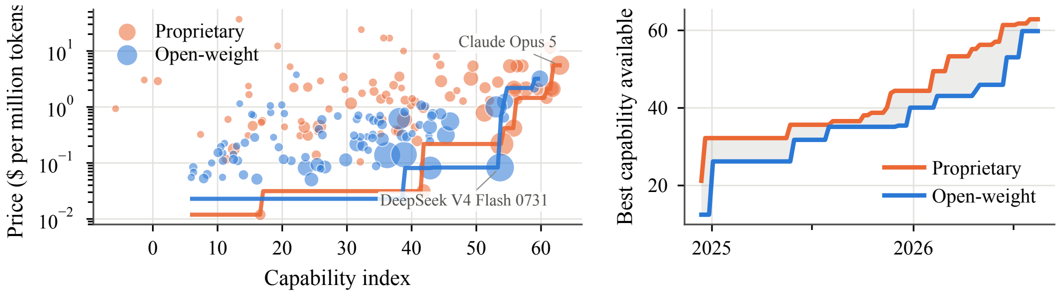

Figure 1: Left, posted blended price against the capability index, marker area proportional to the square root of token volume, step functions giving the cheapest price available at or above each capability level. Right, the best capability available from each side by release date, shading the open-weight lag.

3.2 Volume shares and revenue shares

Open-weight models account for 68% of tokens but only 30% of spending, and the wedge is visible provider by provider. Figure 2 shows the two distributions side by side. DeepSeek serves 24% of tokens and collects 6% of spending, while Anthropic serves 7% of tokens and collects 45%. The provider Herfindahl index is 1183 computed on tokens and 2439 computed on spending. By volume the market is moderately concentrated. By revenue it is concentrated. Volume shares describe who runs the world's inference, revenue shares who captures its value.

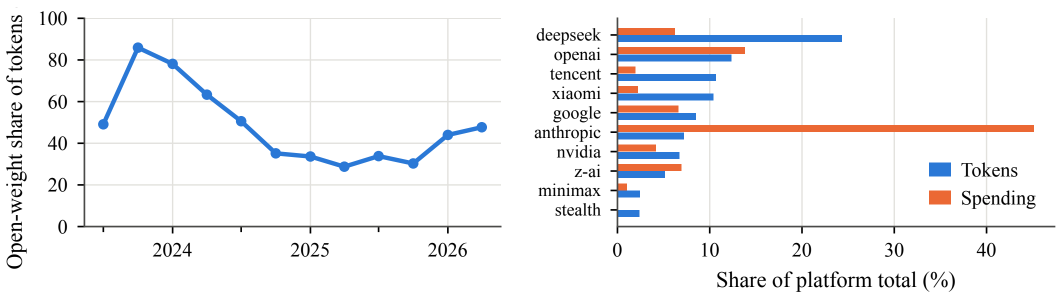

Figure 2: Left, open-weight share of tokens by quarter, computed from the archived leaderboards. Right, share of platform tokens and of spending for the ten largest providers by volume.

The open-weight share of volume is also far from stable. In the archived leaderboards, open-weight models held 86% of tokens in late 2023, fell to 29% by mid-2025, and recovered to 48% by mid-2026. The trough coincides with the sample used by Nagle and Yue [2025], whose finding that proprietary models dominate usage describes a market that has since moved.

3.3 Supply of open weights

Openness also changes who sells the model. Weighting by volume, an open-weight model is served by 14.9 independent hosting providers against 3.0 for a proprietary model, and only 4% of open-weight volume runs on a model with a single host, against 12% of proprietary volume (Figure 7 in the appendix). The price a user pays for an open model is set by hosts competing to serve the same weights rather than by the organisation that trained them, though among multi-host models the ninetieth-percentile host still charges about 1.6 times the cheapest.

4 Price setting and demand

4.1 Frequency of price revision

The archived catalogues let us follow the posted price of a model version from its first appearance. Between consecutive captures, a median of three days apart, the price changes 2.9% of the time, and 54% of models never change price at all. The left panel of Figure 3 shows 56% of models still at their launch price after six months and 45% after a year. Conditional on a change the median revision is a cut of about 5 log points, and 58% of changes are cuts.

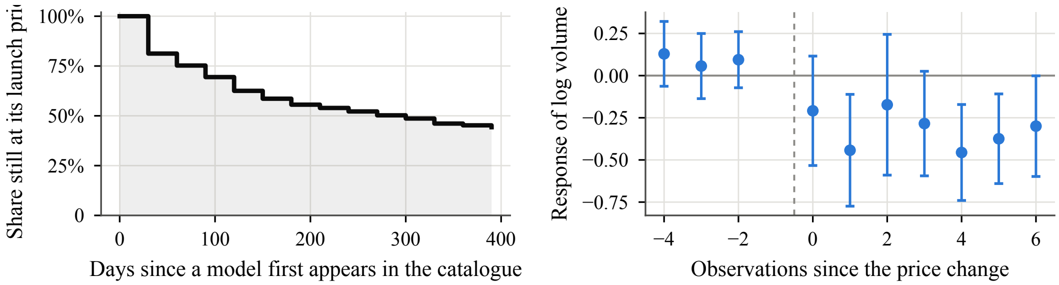

Figure 3: Left, Kaplan-Meier survival of a model's launch price. Right, coefficients β_τ from (5) around 306 revisions, 95% intervals clustered by model.

Expressed the way the price-setting literature does, 12.6% of model-months contain a price change, an expected duration of about 7.4 months. That is at the low end of the range for consumer goods and comparable to online retail [Bils and Klenow, 2004, Nakamura and Steinsson, 2008, Gorodnichenko and Talavera, 2017, Cavallo, 2018], and that comparison is itself the surprise. The menu cost that explains rigidity in retail is close to zero when the price is a field in an API catalogue, and the managerial-attention frictions estimated for online sellers [Ellison et al., 2018] bind far less on a provider tracking a handful of models than on a retailer tracking thousands. What is missing is not the ability to reprice but the incentive. Over the few weeks a release event study spans, an incumbent's posted price is unchanged with probability near 0.9, so a design that looks for the competitive effect of open weights in incumbent price series is looking at a variable that rarely moves.

4.2 Volume response to price revisions

We use the rare revisions to estimate how volume responds. Merging the archived leaderboards with the archived catalogues gives 19,803 model-date observations on 634 models between November 2023 and May 2026, containing 708 observations that follow a price change. We regress log volume on the log fixed-weight price with model and date fixed effects, so that identification comes only from within-model revisions, net of anything common to the market on a date,

$$\log y_{jt} = \beta \log p_{jt} + \alpha_j + \gamma_t + \varepsilon_{jt}. \tag{4}$$

Here α_j is a model effect, γ_t a date effect and ε_{jt} the residual. We estimate $\hat{\beta} = -0.53$ (s.e. 0.10) for tokens and -0.46 (s.e. 0.09) for requests, with indistinguishable estimates for open-weight and proprietary models. Dropping the model fixed effects, so that cross-model variation identifies β, moves the estimate to -0.24, which is what one would expect if higher-priced models carry unobserved quality. Figure 4 shows that the estimate survives every alternative we tried, holding between -0.66 and -0.46 across ten specifications and between -0.58 and -0.51 when models are dropped one at a time.

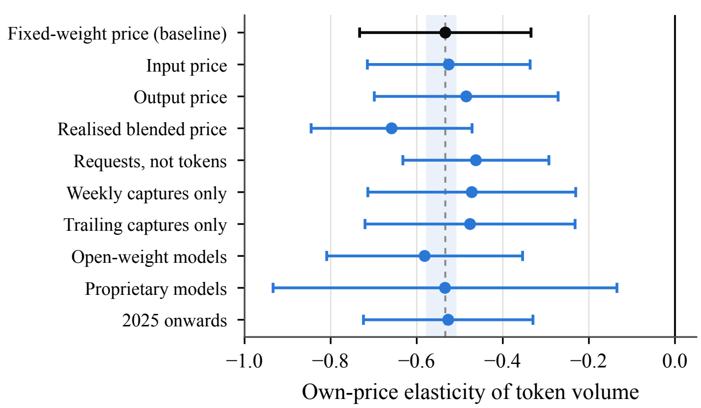

Figure 4: Own-price elasticity across specifications, with 95% intervals clustered by model. Shading marks the range spanned by dropping each of the 268 models whose price ever moves, one at a time.

The right panel of Figure 3 traces the dynamics around 306 revisions, restricted to the captures that report non-overlapping weekly buckets so that the path is not an artefact of a trailing aggregation window. The specification replaces the level term in (4) by leads and lags interacted with the size of the revision Δ_{jk},

$$\log y_{jt} = \sum_{\tau \neq -1} \beta_\tau \, \Delta_{jk} \, \mathbf{1}\{t - T_{jk} = \tau\} + \delta_{jk} + \gamma_t + \varepsilon_{jt}, \tag{5}$$

where T_{jk} is the date of revision k to model j and δ_{jk} a revision-specific effect. Volume is flat before the change and settles about 0.3 to 0.45 log points lower per log point of price increase within a few weeks. The pre-period coefficients are small and individually insignificant, which is the relevant check. Had providers raised prices after volume rose, the estimate would be biased toward zero, and we find no such pattern.

An elasticity near one-half is not consistent with a static Bertrand equilibrium in posted prices, which would require demand to be elastic at the optimum. We read it as a short-run object. Switching a production workload to another model requires re-testing prompts, harnesses and evaluation suites, the kind of switching cost that sustains price dispersion in equilibrium [Klemperer, 1995, Farrell and Klemperer, 2007], so the relevant margin adjusts over months rather than days. That also rationalises the rigidity. A provider that would gain little volume from a price cut, and lose little from holding steady, has weak incentive to reprice a model already in the field.

5 Model entry

5.1 Decomposition of volume growth

The daily panel covers every model with usage for the four weeks to 20 August 2026, a window containing 10 releases by providers with priced, scored models, of which 5 are open-weight, and over which the market grew 65%. The left panel of Figure 5 decomposes daily volume into incumbents and models released during the window. Entrants reached 28% of platform tokens within four weeks, 69% of it open-weight. Incumbent volume did not fall. It grew 18%. Models released during the window account for 72% of net growth over it, and incumbent expansion for the rest.

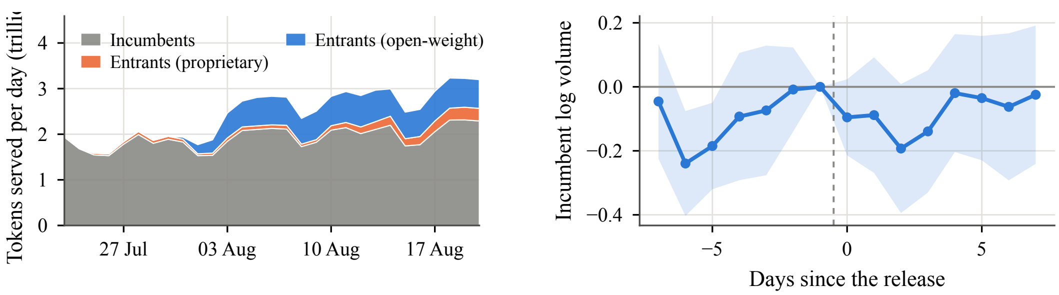

Figure 5: Left, daily platform volume split between models already serving traffic at the start of the window and models released during it. Right, event-time coefficients from (6) per unit of exposure (7), 95% intervals clustered by incumbent.

5.2 Incumbent responses to entry

Aggregate volume can hide reallocation, so we test for it directly. For each release k at date T_k we stack all incumbents j that were already serving traffic, and estimate

$$\log y_{jt} = \beta \, \mathbf{1}\{t \geq T_k\} \times \tau_{jk} + \delta_{jk} + \lambda_{kt} + X'_{jt}\theta + \varepsilon_{jkt}, \tag{6}$$

where δ_{jk} is an incumbent-by-entrant effect and λ_{kt} an entrant-by-day effect, so that β compares incumbents exposed to the same release on the same day. The exposure measure combines technological proximity with the entrant's relative value for money,

$$\tau_{jk} = S_{jk} \, r_{jk}, \qquad r_{jk} = \log \frac{Q_k}{p_k} - \log \frac{Q_j}{p_j}, \tag{7}$$

with S_{jk} from (2). Because capability and price both trend, X_{jt} also contains post-release interactions with the incumbent's own capability and price.

We find no diversion. The estimate of β is 0.01 (s.e. 0.07), and the event-time path in the right panel of Figure 5 is flat on both sides of the release. The confidence interval excludes differential declines larger than 0.13 log points per unit of exposure, tightening to 0.10 and 0.04 under the Euclidean and cosine measures in Table 4. Ten releases bound the effect rather than pin it down, but at this rate of growth the displacement channel is plainly small relative to expansion.

5.3 Longer-horizon estimates

The archived panel lets us look further out. We take the 19 open-weight releases that reached at least one percent of platform tokens and compare aggregate volume of models already in the market over the eight weekly observations before and after each. Incumbent volume fell in 32% of these episodes, and the median change was $+0.25$ log points against a median market change of $+0.52$. DeepSeek R1 [DeepSeek-AI, 2025] is the sharpest case, with platform volume rising 0.56 log points over the sixteen weeks around it while incumbent volume held flat at -0.03. Incumbents lose share because the market grows around them, not because they lose volume.

6 Demand system and counterfactuals

6.1 Specification

Because posted prices do not respond and incumbents are not displaced, the value of open-weight competition is measured on the choice set rather than on prices. We estimate a nested-logit demand system over the 172 models with prices and capability scores, nesting by capability tercile so that models of similar capability are closer substitutes. Writing $g(j)$ for the nest of model j and $\sigma \in [0, 1)$ for the within-nest correlation [Cardell, 1997, Berry, 1994], shares take the form

$$s_j = \frac{\exp\big(\delta_j/(1-\sigma)\big)}{D_{g(j)}} \cdot \frac{D_{g(j)}^{1-\sigma}}{1+\sum_h D_h^{1-\sigma}}, \qquad D_g = \sum_{k \in g} \exp\big(\delta_k/(1-\sigma)\big), \tag{8}$$

with mean utility

$$\delta_j = \beta_p \log p_j + \alpha' x_j + \xi_j, \tag{9}$$

where x_j collects the capability index, context length, modality, reasoning support, latency and age, α the coefficients on them and ξ_j the unobserved quality of model j. The outside option is the unscored fringe, whose share s_0 is 2.4% of tokens, and $s_{j|g(j)}$ denotes model j's share within its own nest. Shares invert to mean utilities [Berry, 1994],

$$\delta_j = \log s_j - \log s_0 - \sigma \log s_{j|g(j)}, \tag{10}$$

so the system reproduces the data exactly and the counterfactual does not turn on how well the characteristics fit. Consumer surplus follows from the expected maximum utility [Small and Rosen, 1981, McFadden, 1974],

$$\text{CS} = \frac{1}{|\beta_p|} \log\Big(1 + \sum_h D_h^{1-\sigma}\Big). \tag{11}$$

Deleting the open-weight models changes the denominators in (8), redistributing their volume across the remaining models and the outside option, so the counterfactual depends only on (β_p, σ).

6.2 Parameter choice

Cross-sectional share variation does not identify β_p and σ here. The differentiation instruments of Gandhi and Houde [2019] are functions of the distribution of rival characteristics, which in this market is a distribution over an essentially one-dimensional index, so they are near-collinear with a model's own capability and with its unobserved quality. Parameter count, the obvious cost shifter, correlates 0.65 with the capability index and moves the price coefficient from -0.79 under least squares to -0.27 despite a first-stage F above 900, a violated exclusion restriction rather than a corrected bias.

We therefore fix $\beta_p = -1$ and report every result across a range of σ [Nevo, 2000, Conlon and Gortmaker, 2020]. Both choices are conservative. The price coefficient exceeds in magnitude the within-model estimate of Section 4.2 and the long-run elasticity of about -0.6 implied by (12), and because surplus enters (11) divided by $|\beta_p|$ a larger magnitude yields a smaller dollar figure. Finding no differential loss within the entrant's own capability tier is what σ near zero predicts, and the surplus loss is largest there, so $\sigma = 0.4$ is conservative too.

6.3 Removing open-weight models

Removing open-weight models from the choice set at posted prices, the average price paid per token rises from \$0.70 to \$1.80, an increase of 156%. Consumer surplus (11) falls 21%, worth about \$2.4m per day at current volume. The percentage change does not depend on β_p, which enters the baseline and the counterfactual identically, and only the conversion into dollars scales with it. The loss runs from \$3.9m to \$1.3m per day as σ moves from 0 to 0.7 (Table 5), and raising the outside share from the observed 2.4% to 50% raises the proportional loss from 21% to 47% while lowering the dollar figure to \$1.0m per day. The price effect is invariant to both, because it is governed by the composition of the surviving choice set rather than by the strength of substitution. Concentration moves against the presumption that open weights fragment the market. the provider Herfindahl index on tokens rises from 897 to 1048, because open-weight volume is spread across more suppliers than the proprietary volume that replaces it.

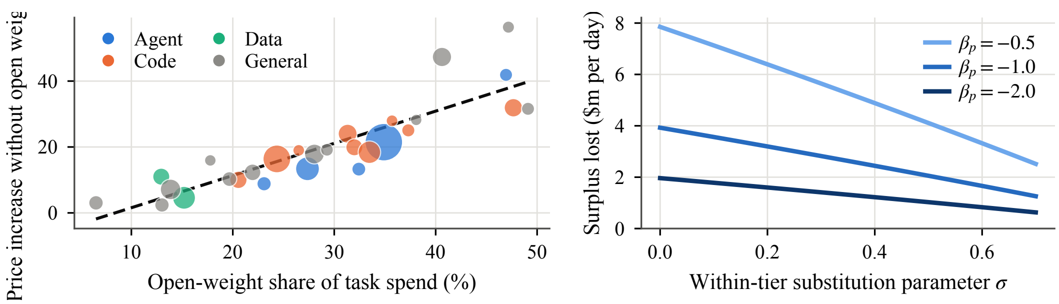

Figure 6: Left, for each of the 29 task markets, the price increase implied by removing open-weight models against their share of task spending, marker area proportional to the task's share of platform spending. Right, aggregate surplus loss per day across σ and β_p.

These numbers hold prices fixed, which given the rigidity of Section 4.1 makes the exercise a statement about variety in the tradition of Trajtenberg [1989], Hausman [1996] and Petrin [2002]. Allowing providers to reprice would add to the loss, so the estimates are a lower bound.

The left panel of Figure 6 repeats the exercise separately in each of the 29 task-level markets, where shares of spending are reported directly. The implied price increase rises almost one for one with the open-weight share of task spending, with a slope of 0.98. Weighted by spending, prices rise 18% and surplus falls 13%, with the largest effects in agentic and coding work, where open-weight models hold about a third of spending, and the smallest in data extraction and transformation, where they hold under a sixth. These are smaller than the aggregate because task shares are measured on spending, which weights the expensive proprietary models that open weights do not displace. Table 2 groups the tasks into families. The effect tracks open-weight penetration closely, reaching 20% in coding work and 7% in data extraction and transformation.

Table 2: The open-weight counterfactual inside each family of task markets, weighted by each task's share of platform spending. Shares and changes are averages over the tasks in the family.

Task family	Tasks	Share of spending	Open-weight share	Price without open
Agent	5	0.29	0.33	+19%
Code	9	0.30	0.31	+20%
General	12	0.22	0.25	+19%
Data	2	0.09	0.15	+7%

7 Interpretation

7.1 An accounting framework

The three results fit together through the share of demand that is in play. Let M_t be total demand growing at rate g per period, and let a fraction ϕ of existing demand be revisited each period as workloads are rebuilt or re-evaluated. Contestable demand is $(g + \phi)M_t$, and a model's volume evolves as

$$y_{j,t+1} = (1 - \phi)\, y_{jt} + \nu_j\, (g + \phi)\, M_t, \tag{12}$$

where ν_j is its share of contestable demand. An incumbent loses volume only when $\nu_j(g+\phi)M_t < \phi\, y_{jt}$, so with $g \approx 0.13$ per week in our window an incumbent holding an average share of new demand keeps growing. An entrant reaches a share of about $\nu_k(g + \phi)/(1 + g)$ after one period, taking volume quickly without anyone shrinking. And a permanent price change reaches only the contestable margin, so the response measured τ periods out is attenuated by roughly $1-(1-\phi-g)^\tau$ relative to its long-run value.

That last point bounds how much of the inelasticity is an artefact of slow adjustment. With $\phi + g \approx 0.23$ per week and the six weekly observations over which Figure 3 shows the response settling, the attenuation factor is about 0.8 and the implied long-run elasticity is near -0.6. Demand is inelastic even after adjustment, so the first-order condition of a provider maximising current profit is not satisfied at the posted price. Either providers price against a margin we do not observe, or the posted price is not chosen to maximise current profit at all, and the rigidity of Section 4.1 favours the second.

The foregone savings Nagle and Yue [2025] identify are a puzzle if demand adjusts quickly and are exactly what one should expect when the short-run elasticity is about one-half and switching a workload is costly. They accrue as workloads turn over rather than as incumbents reprice. Our price-of-capability gaps are smaller than theirs, consistent with proprietary providers having released cheaper tiers since and with the rebound in the open-weight volume share in Figure 2.

7.2 Limitations

Three qualifications bound what the estimates cover. The market is one router that selects for price-sensitive developers, so the estimates describe the segment where substitution toward open weights is easiest. The demand parameters are calibrated rather than estimated, which is why every counterfactual is reported over a range and at conservative values. And four weeks of releases support a bound on diversion rather than a point estimate, since the staggered-entry estimators that handle heterogeneous treatment timing [Callaway and Sant'Anna, 2021, de Chaisemartin and D'Haultfœuille, 2020, Sun and Abraham, 2021, Borusyak et al., 2024] need more events than the window provides.

8 Conclusion

Open-weight models constrain proprietary providers on a margin the usual measurement misses. Posted prices behave like a fixed characteristic of a model version. More than half of models are never repriced, and the monthly frequency of a change is comparable to groceries in a setting where changing the number costs nothing. Demand explains why. The within-model elasticity is about one-half, and correcting for slow reallocation moves it only to -0.6, so a provider gains little volume from cutting the price of a model already in the field and does not cut it. Entry is absorbed by growth. Over four weeks, models released inside the window took more than a quarter of platform tokens while incumbent volume rose, and we find no differential loss for the incumbents nearest an entrant. Across nineteen major open-weight releases since 2024, incumbent volume fell in fewer than a third of episodes.

The constraint is nonetheless large. Removing open-weight models from an estimated demand system raises the average price paid per token by more than half again and costs users roughly a fifth of their surplus, on the order of a million dollars a day on one platform, with the largest effects in agentic and coding work. It also raises provider concentration rather than lowering it, because open-weight volume is spread across more suppliers than the proprietary volume that would replace it. An assessment built on incumbent price responses will therefore find nothing, not because the constraint is absent but because it operates on the price at which the next unit of demand is served.

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A Data construction

Endpoints. The catalogue endpoint returns, for every listed model, the posted input and output price per token, context length, input and output modalities, a creation timestamp, and the Hugging Face repository identifier when the weights are public. The leaderboard endpoint returns tokens, requests, tool calls and media counts by model and serving variant. A benchmark endpoint returns third-party capability scores and usage-weighted effective prices. A task endpoint returns, for each of 29 usage categories, the category's share of platform spending and the share held by each of its ten largest models, in both spending and token terms. An endpoint listing gives each hosting provider's price, quantisation and uptime for a given model. All are public and unauthenticated.

Daily usage. Each model's public page embeds its own trailing-month daily usage series in the page's streamed render payload. We request one page per model with observed usage and parse the payload, which yields 457 models over 30 days. The same payload carries per-endpoint latency and throughput percentiles, which we aggregate to the model level by request count.

Historical coverage. Internet Archive captures supply history. Captures of the catalogue endpoint give a price and characteristic panel from July 2023. Captures of the leaderboard page embed the leaderboard payload in the same streamed format, so we recover model-level usage without relying on the rendered page. Two vintages are present and we treat them separately: captures before late 2024 embed a series of non-overlapping weekly buckets, and later captures embed a single trailing-window snapshot. Shares are comparable across vintages; levels are not, so all specifications include date fixed effects, and Table 3 reports the two vintages separately. We fetch captures at a fixed spacing with adaptive backoff, which is the binding cost of the collection: the archive requests take a few hours, and every other step runs in minutes on a laptop.

Merging. Usage and price observations are matched by model identifier and nearest date within ten days. Openness is taken from the current catalogue where the model is still listed, because the weight-repository field is absent from the earliest captures. Prices are dollars per million tokens throughout.

B Capability imputation

Benchmark scores are missing for smaller models. For each missing dimension we fit a ridge regression on the dimensions the model does have, using models scored on both, and predict the missing value. Masking half of the fully scored models and predicting the held-out values gives R^2 of 0.99, 0.97 and 0.96 for the reasoning, coding and agentic dimensions, with mean absolute errors of 1.7, 5.0 and 3.1 points against standard deviations of 16.4, 22.6 and 18.5. Imputation raises coverage from 109 models and 77% of tokens to 180 models and 91% of tokens. All results in the main text hold on the directly scored subsample.

C Robustness

Table 3 reports the elasticity across price measures, outcome measures, capture vintages, openness and sample period. The estimate stays between -0.38 and -0.66. The realised blended price, which mixes posted prices with the model's own input-output composition, gives the largest magnitude, as expected if a shift toward prompt-heavy workloads lowers the average price per token and raises volume at the same time; the fixed-weight index removes that channel and is our preferred measure.

The estimate is not driven by any one model: dropping each of the 268 models whose price ever moves, one at a time, leaves it between -0.58 and -0.51. Restricting to models holding at least 0.01% of tokens gives -0.378 (s.e. 0.127).

Two further checks bear on the elasticity. A price cut may accompany the launch of a replacement by the same provider, which would depress the old model's volume for reasons other than price. Adding an indicator for observations within 28 days of a same-provider release, which covers 38% of the panel, leaves the estimate at -0.534; dropping those observations gives -0.499 (s.e. 0.123).

Table 4 reports the diversion test under three definitions of technological distance. None of the six estimates is distinguishable from zero. The Euclidean and cosine measures give tighter standard errors than the Mahalanobis measure, and their confidence intervals bound the differential decline for the most exposed incumbents at 0.10 and 0.04 log points respectively.

Table 3: Own-price elasticity across specifications. Each row is a separate estimate of (4) with model and date fixed effects; standard errors clustered by model.

Specification	Estimate	Std. error	Observations	Models
Fixed-weight price (baseline)	-0.534	0.102	19,803	634
Input price	-0.526	0.096	19,803	634
Output price	-0.485	0.109	19,803	634
Realised blended price	-0.658	0.095	19,803	634
Requests rather than tokens	-0.463	0.087	19,803	634
Weekly captures only	-0.472	0.123	15,133	432
Trailing captures only	-0.476	0.125	4,670	428
Open-weight models	-0.582	0.116	8,873	321
Proprietary models	-0.534	0.204	10,930	316
2025 onwards	-0.527	0.100	7,906	542

Table 4: Diversion at entry under alternative distance measures. Each row estimates (6) with incumbent-by-entrant and entrant-by-day fixed effects and post-release controls for the incumbent's own capability and price.

Distance	Exposure	Estimate	Std. error	Observations
Mahalanobis	similarity $\times$ relative value	0.010	0.070	11,820
Mahalanobis	similarity	0.019	0.120	11,820
Euclidean	similarity $\times$ relative value	-0.006	0.047	11,820
Euclidean	similarity	0.008	0.065	11,820
Cosine	similarity $\times$ relative value	0.022	0.031	11,820
Cosine	similarity	0.083	0.086	11,820

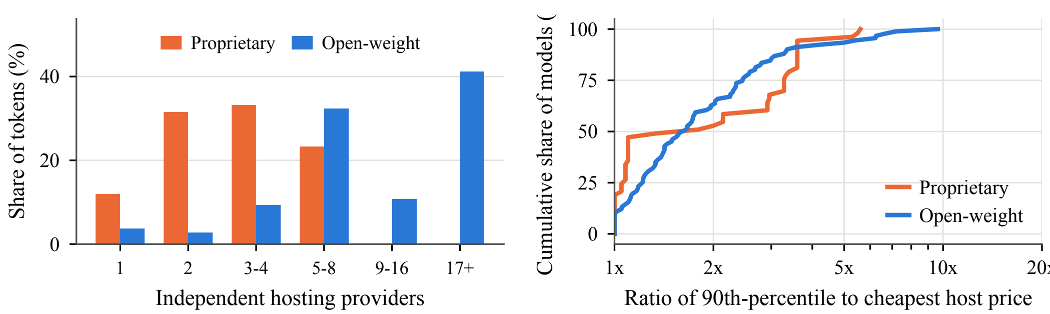

Figure 7: Left: distribution of tokens across the number of independent hosting providers serving the model. Right: cumulative distribution, across models with more than one host, of the ratio between the ninetieth-percentile and the cheapest posted host price.

D How good and how cheap an open model has to be

The counterfactual removes the open-weight models that exist. The opposite exercise asks what a new one would have to look like to matter. We insert a single open-weight entrant with capability $(1-\rho)\max_k Q_k$ and price p, assign it the mean utility implied by the fitted gradient of (9) in capability and log price, and recompute shares. Figure 8 traces the effect on proprietary volume. The iso-effect lines are steep, which is the substantive point: price moves the outcome far more than the last few points of capability. An entrant that matches the frontier exactly but is priced at \$0.50 per million tokens takes 6.6% of platform tokens and cuts proprietary volume by 7%. An entrant a tenth below the frontier but priced at \$0.05 takes 17% and cuts proprietary volume by 18%. At the frontier's own price, an open model that matches it takes essentially nothing.

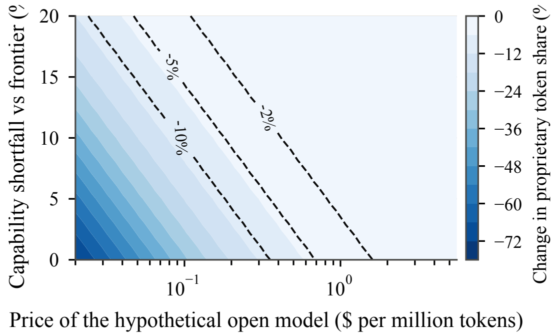

Figure 8: Change in proprietary token share when one hypothetical open-weight model is added to the choice set, by its price and its capability shortfall against the frontier, at $\beta_p = -1$ and $\sigma = 0.4$. Contours mark 2, 5 and 10 percent losses.

E Counterfactual sensitivity

Table 5 reports the counterfactual across the within-tier substitution parameter, and `robustness.json` reports it across the size of the outside option as well.

Table 5: The open-weight counterfactual across substitution parameters, at $\beta_p = -1$. Prices are dollars per million tokens; the surplus loss is evaluated at the platform's volume on the reference day.

σ	Price paid ($/M)	Price without open ($/M)	Surplus change (%)	Surplus loss ($m per day)	HHI (without open)
0.0	0.70	1.82	−33.8	3.9	1109
0.2	0.70	1.81	−27.5	3.2	1082
0.4	0.70	1.80	−21.0	2.4	1048
0.6	0.70	1.79	−14.3	1.7	1006
0.7	0.70	1.78	−10.8	1.3	982

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